

MUHAMMED MAZHER KHAN

AI Engineer (LLM Specialization) | LangChain • PyTorch • Hugging Face

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SUMMARY

LLM-focused AI Engineer with 2+ years of production experience designing, fine-tuning, and deploying Hugging Face transformers and OpenAI-powered agents into scalable web applications. Specialist in agentic workflows using LangChain/LangGraph, RAG architectures, structured tool calling, and context management (functional equivalent of MCP). Proven track record of integrating LLMs with real-time data pipelines (Kafka, AWS Glue/Lambda), vector retrieval, and Kubernetes-orchestrated services at Deloitte. Full-stack background (Django/Flask/FastAPI + React) enables end-to-end ownership from model selection and prompt engineering to API design and cloud deployment.

Experience

Solution Analyst

Deloitte Consulting LLP

06/2023 - 01/2025, Mechanicsburg, PA

A leading global consulting firm known for providing IT, audit, consulting, tax, and advisory services.

Project 1:

- Developed and optimized data pipelines to preprocess and feed structured datasets into Hugging Face transformer models, improving NLP task performance by 15%.
- Integrated OpenAI API to dynamically generate and refine prompts, enhancing conversational AI feature quality by 20% in a web application.
- Designed RESTful APIs using Flask and Django REST Framework, integrating Hugging Face and OpenAI models with JavaScript (TensorFlow.js) front-ends for real-time user interactions.
- Implemented client-side JavaScript with TensorFlow.js to preprocess user inputs, reducing server load by 10% while maintaining model accuracy.
- Orchestrated scalable deployment of containerized applications on AWS using Kubernetes, Terraform, and CI/CD pipelines with GitHub Actions and AWS CodePipeline.
- Built and optimized data pipelines with AWS Glue, AWS Lambda, and Apache Kafka for real-time data streaming and batch processing, improving throughput by 25%.
- Implemented agentic workflows and structured tool calling using LangChain/LangGraph (functionally equivalent to MCP) for dynamic retrieval and multi-step reasoning.
- Integrated embeddings with FAISS/Pinecone for RAG-based question answering and document retrieval.
- Implemented unit tests with Python's unittest and pytest, achieving 90% code coverage, and executed SQL queries to validate data integrity for testing.

Key Achievements

API Development Excellence

Developed secure RESTful APIs with 98% uptime for client applications.

Blockchain Innovation

Implemented blockchain solutions with 20 smart contracts in real-world projects.

Efficient Deployment Strategies

Automated deployment pipelines, reducing deployment time by 30%.

Batch Processing Optimization

Led initiatives improving batch processing speeds by 40% with Databricks.

SKILLS

Agile Python Algorithms C# VB

API Development TorchServe ReactJs

AJAX ASP.NET REST API TensorFlow

GCP Azure PyTorch scikit-learn

Hugging Face Transformers TensorFlow.js

LangChain LangGraph (MCP-equivalent context/tool protocols) Big Data Nodejs

TFS CI/CD Github Actions

Cloud Computing SQL OAuth2

Javascript Typescript xUnit JQuery

Configuration CSS Pandas CSV
CUDA

Data Extraction data integrity OpenAI API

data migration Pydantic Data Quality

- Designed monitoring and alerting systems using AWS CloudWatch for batch processing jobs, reducing error detection time by 30%.

Project 2:

- Enhanced internal ASP classes to ensure HTML5 generation adhered to ADA compliance standards, incorporating responsive design principles for web and mobile accessibility.
- Developed web applications using ASP.NET MVC, Web API, and WCF frameworks, integrating with MS SQL Server.
- Facilitated daily scrum meetings to ensure adherence to Agile best practices.
- Implemented Apache Kafka for real-time data streaming and messaging.
- Deployed and managed containerized applications using Kubernetes.
- Utilized Databricks and Azure Function Apps to implement scalable data pipelines.
- Conducted root cause analysis to identify and resolve system errors.
- Collaborated with design team for class, sequence diagrams, and UX wireframes to align engineering with user experience goals.
- Optimized existing AJAX files and leveraged jQuery, Bootstrap, and SASS for enhanced front-end development.
- Designed and optimized SQL stored procedures, created views, and improved database performance.
- Implemented unit tests using xUnit.
- Executed SQL queries to retrieve precise data for testing.
- Innovated unit testing methodologies for ADA compliance.
- Orchestrated deployment of application updates to Azure Cloud using Infrastructure as Code (IaC) principles.
- Collaborated on automated deployment pipelines with GitLab and GitHub Actions.
- Managed batch processing operations, conducting performance analysis and optimization.
- Implemented monitoring and alerting systems for batch processing jobs.
- Provided technical support during batch processing and deployment operations.
- Designed and implemented CI/CD pipelines.
- Documented workflows and procedures, including style guides for consistent UI development.

Director/ I.T Consultant

Mansalvic Consulting LLP

01/2023 - 03/2023, Cleveland, OH

A software IT consulting firm focused on providing IT services and software solutions.

Lead Marketing campaign to reach out to potential client explaining our services. And provide consultation on software matters

- Check emails, messages, or free communication tools (e.g., Slack, WhatsApp) for client updates or urgent issues.

deployment pipelines

DigitalOcean Docker Flask MongoDB

Git Github NodeJs

HTML IIS

Training

Artificial Intelligence/ Machine Learning

Courses focused on the principles of Artificial Intelligence and Machine Learning.

Projects

Github link:

[https://github.com/Muhanned7?](https://github.com/Muhanned7?tab=repositories)

[tab=repositories](#)

Text Classifier

1/2023 - 09/2023

A project focused on building robust data extraction pipelines from PDF files.

- Developed robust data pipelines to extract raw data from PDFs.
- Leveraged regex to identify and extract patterns from PDF text.
- Transformed unstructured data into structured formats for analysis.
- Applied domain-specific rules to enrich the dataset.
- Automated the pipeline for scheduled data extraction and cleaning.

Node.js Backend Service (Personal/Professional)

Designed scalable RESTful APIs using Node.js and Express, integrated with MongoDB/Redis for caching and PostgreSQL for relational data. Implemented data modeling, indexing, and query optimization for high-performance retrieval.

Neural Network Implementation

Developed a neural network for performance optimization in machine learning.

- Developed robust data pipelines to extract raw data from PDFs.
- Leveraged regex to identify and extract patterns from PDF text.
- Transformed unstructured data into structured formats for analysis.
- Applied domain-specific rules to enrich the

- Review progress on active client projects using free tools like Google Sheets, Notion, or Trello.
- Prioritize tasks for the day, balancing client deliverables with efforts to find new work.
- Writing code, configuring systems, or setting up IT solutions (e.g., websites, small apps, or network setups).
- Troubleshooting issues like website bugs, server errors, or basic cybersecurity setups.
- Preparing deliverables (e.g., reports, documentation) for clients.

Research Assistant, BlockChain Development

Cleveland State University

01/2022 - 08/2022, Cleveland, OH

A public research university that focuses on comprehensive higher education and public service.

Worked on implementation and development of blockchain technologies for experimental purposes.

- Developed smart contracts using Remix and stored node data using MS SQL Server.
- Used Brownie to test smart contract frameworks and worked extensively with Web3 and NodeJS.
- Developed localhost operations for testing using Python and Typescript.
- Performed efficiency tests on mining algorithms and analyzed performance.
- Plotted data and presented findings on mining algorithms

Education

Master of Science, Computer Science

Cleveland State University

01/2020 - 12/2022

Bachelor of Technology, Mechanical Engineering

JB Institute of Engineering and Technology

11/2010 - 11/2015

dataset.

- Automated the pipeline for scheduled data extraction and cleaning.

API Development

A backend project utilizing FastAPI to create a RESTful API.

- Built a RESTful API with FastAPI and MS SQL Server, deployed on Azure.
- Implemented core backend functionalities and SQL Server integration.
- Utilized Pydantic to define and validate data inputs.
- Deployed the application on DigitalOcean.

Blockchain Development

Comprehensive blockchain development focusing on secure user authentication.

- Developed smart contracts for secure data storage and retrieval in a blockchain environment.
- Implemented Web3 technology to enhance user interaction.
- Conducted tests on multiple local blockchain instances.

Cloud Computing

A cloud computing project for processing time log files.

- Designed a Python-based desktop application for processing time logs, enhanced with a mobile-responsive UI using Angular 14 and Bootstrap.
- Deployed the application on a cloud platform for efficient processing.
- Developed an intuitive user interface for user interaction.

Distributed Application

A project focusing on creating a distributed application leveraging RPC.

- Created a distributed application using Remote Procedure Call (RPC) and CUDA.
- Created a distributed application using RPC and CUDA, with a WCF-based desktop client.
- Designed a fault-tolerant system utilizing ONC RPC protocol.

